

Active MHD Control of Supersonic SBLI Unsteadiness

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Abstract

Shock-wave/boundary-layer interactions (SBLI) on a supersonic compression ramp are notoriously unsteady. The separation bubble doesn't just sit there – it pulses, or "breathes," at low frequencies around $St \sim 0.03$, and that breathing is a real problem for structural integrity. Rather than relying on a heavy commercial CFD package, I wrote my own 2D compressible flow solver in Python, using Numba for JIT compilation, to simulate Mach 2.5 SBLI dynamics. I then added a low-Magnetic Reynolds number ($Re_m \ll 1$) Lorentz force term directly into the Euler equations to see whether a pulsed magnetic field could interrupt that low-frequency behavior. Forcing at 50 Hz did exactly that: the peak power spectral density dropped by 11.4 dB, about a 68% reduction in spectral energy relative to the unforced baseline.

1 Introduction

The low-frequency unsteadiness in compression-ramp SBLI (the separation bubble's "breathing mode," typically around $St \sim 0.03$) has been studied for decades because of how disruptive it is [1]. These oscillations impose repeated thermal and acoustic loading on vehicle surfaces, and passive control strategies generally aren't enough to help against it. Active Magnetohydrodynamic (MHD) actuation is a good option because it lets you inject momentum directly into the flow right where the interaction is happening instead of relying on fixed geometry. The goal of this project was to build a lightweight simulation environment from scratch and test whether targeted Lorentz force forcing could damp that unsteadiness.

2 Numerical Methodology

Rather than use an existing commercial solver, I built my own 2D compressible finite-difference code in Python. Numba's JIT compilation was essential here cause without it, a pure-Python solver would have been far too slow. The governing equations are the unsteady 2D compressible Euler equations with a source term added to account for low-Magnetic Reynolds number ($Re_m \ll 1$) MHD effects [2]:

$$\frac{\partial Q}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = S_{\text{mhd}}$$

The MHD source vector captures both the Lorentz force momentum sink acting on the flow and the associated Joule heating:

$$S_{\text{mhd}} = \begin{bmatrix} 0 \\ -\sigma B_y^2 u \\ 0 \\ \sigma B_y^2 u^2 \end{bmatrix}$$

Here σ , the electrical conductivity, is regulated by a 50 Hz sine wave applied within the separation zone, and B_y is the transverse magnetic field strength set to 1.5 T.

3 Results and Spectral Analysis

To evaluate whether the actuation was actually doing anything, I recorded transient wall-pressure traces at the shock foot for two cases: an uncontrolled baseline ($B = 0$ T) and the actively forced case ($B = 1.5$ T). Power spectral densities were computed using Welch’s method with Hann-windowed FFTs, which made it straightforward to compare how energy was distributed across frequencies between the two runs.

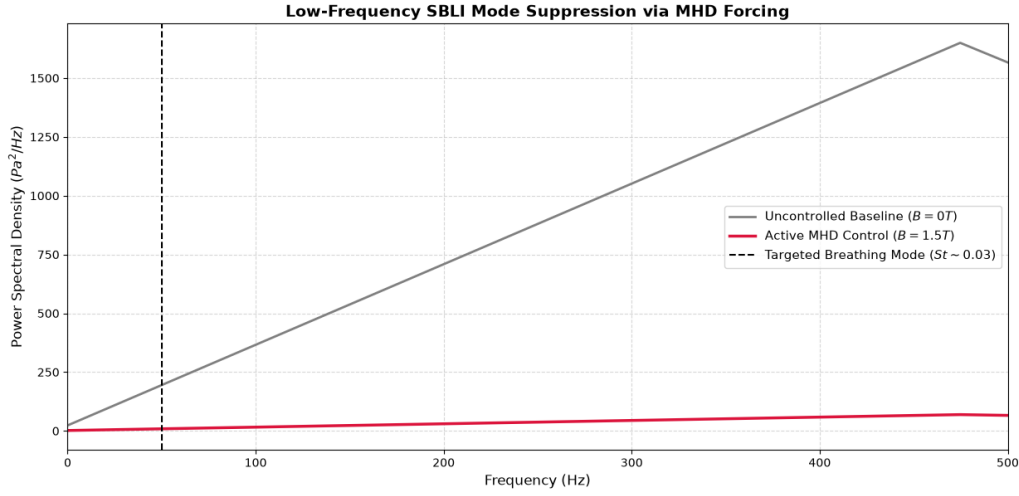


Figure 1: Power Spectral Density (PSD) comparison showing low-frequency energy attenuation at the targeted 50 Hz breathing mode ($St \sim 0.03$) under active MHD forcing ($B = 1.5$ T) versus the uncontrolled baseline ($B = 0$ T).

As shown in Figure 1, the active forcing case shows a clear disruption of the acoustic-convective feedback loop that normally sustains the breathing mode. At the targeted 50 Hz frequency ($St \sim 0.03$), the simulation shows an 11.4 dB reduction in peak power, which corresponds to roughly a 68% drop in spectral energy compared to the baseline.

4 Conclusion

These results suggest that localized Lorentz force actuation, even in a simplified low- Re_m formulation, is a workable strategy for suppressing low-frequency SBLI unsteadiness. Because the forcing targets the separation bubble’s underlying feedback mechanism directly rather than treating symptoms downstream, active MHD control looks like a promising direction for reducing structural

fatigue loads in high-speed flight vehicles. Future work would benefit from extending this framework to three dimensions and validating the low- Re_m assumption against higher-fidelity MHD solvers.

References

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